Supporting Information for: "The Impossibility of Faithful Explanation Under Underdetermination"

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S1 Proofs of All Propositions

S1.1 Proof of Tightness (Main Text Proposition 1)

Proof. We construct three witnesses, one for each achievable pair.

(i) Faithful + Decisive (drop Stable). Set $E = \mathsf{explain}$. Faithfulness: for every θ, $\perp(E(\theta), \mathsf{explain}(\theta)) = \perp(\mathsf{explain}(\theta), \mathsf{explain}(\theta))$, which is false by irreflexivity of $\perp$. Decisiveness: for every θ and h, $\perp(\mathsf{explain}(\theta), h) \Rightarrow \perp(E(\theta), h)$ holds by identity. Stability fails: the Rashomon property provides θ_1, θ_2 with $\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2)$ and $\perp(\mathsf{explain}(\theta_1), \mathsf{explain}(\theta_2))$, so $\mathsf{explain}(\theta_1) \neq \mathsf{explain}(\theta_2)$, hence $E(\theta_1) \neq E(\theta_2)$.

(ii) Faithful + Stable (drop Decisive). Let $c \in \mathcal{H}$ satisfy $\neg \perp(c, \mathsf{explain}(\theta))$ for all θ (a "neutral" or "agnostic" explanation). Set $E(\theta) = c$ for all θ. Faithfulness holds by construction. Stability holds because E is constant. Decisiveness fails because E abstains on every distinction: $\perp(\mathsf{explain}(\theta), h)$ does not imply $\perp(c, h)$ in general.

(iii) Stable + Decisive (drop Faithful). Let $c \in \mathcal{H}$ be "maximally committal": for every θ and h, $\perp(\mathsf{explain}(\theta), h) \Rightarrow \perp(c, h)$. Set $E(\theta) = c$. Stability holds (constant map). Decisiveness holds by construction. Faithfulness fails because c may satisfy $\perp(c, \mathsf{explain}(\theta))$ for some θ—indeed, it must, since it inherits incompatibilities from both sides of the Rashomon witness. □

S1.2 Proof of Necessity (Main Text Proposition 2)

Proof. ($\Rightarrow$) If the Rashomon property holds, the main theorem shows no faithful, stable, and decisive E exists.

($\Leftarrow$) If the Rashomon property fails, then for all θ_1, θ_2 with $\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2)$, $\neg \perp(\mathsf{explain}(\theta_1), \mathsf{explain}(\theta_2))$. Set $E = \mathsf{explain}$.

- **Faithfulness**: $\neg \perp(\mathsf{explain}(\theta), \mathsf{explain}(\theta))$ by irreflexivity.
- **Decisiveness**: $\perp(\mathsf{explain}(\theta), h) \Rightarrow \perp(E(\theta), h)$ by identity.
- **Stability**: Suppose $\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2)$. We need $\mathsf{explain}(\theta_1) = \mathsf{explain}(\theta_2)$. By the negated Rashomon property, their explanations are compatible: $\neg \perp(\mathsf{explain}(\theta_1), \mathsf{explain}(\theta_2))$. Under the additional hypothesis that the native explanation is *injective on incompatibility classes* (i.e., compatible explanations on the same observe-fiber are equal), stability follows. In the Lean formalization, this is captured by the `no_rashomon_implies_all_three` theorem, which assumes the negation of the Rashomon property directly implies equality of native explanations on each fiber.

□

S1.3 Proof of G-Invariance Implies Stability (Main Text Proposition 3)

Proof. Let θ_1, θ_2 with $\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2)$. By transitivity of the G-action on observe-fibers, there exists $g \in G$ with $g \cdot \theta_1 = \theta_2$. By G-invariance, $E(\theta_2) = E(g \cdot \theta_1) = E(\theta_1)$. □

S2 All Nine Instance Details

S2.1 Instance 1: Feature Attribution Under Collinearity

Explanation system. Configuration space $\mathbf{\Theta}$: trained models from a fixed class (GBDT, Lasso, neural networks), indexed by random seed. Observable space Y: input–output functions. Explanation space $\mathcal{H}$: attribution vectors $(\varphi_1, \ldots, \varphi_P) \in \mathbb{R}^P$. Incompatibility: $\varphi \perp \varphi'$ if there exist collinear features j, k with $\varphi_j > \varphi_k$ and $\varphi'_k > \varphi'_j$.

Rashomon property. Collinear features induce a Rashomon set: models that differ only in which feature within a correlated group acts as primary predictor [1, 2].

Quantitative bounds. GBDT: ratio $1/(1-\rho^2)$ diverges as $\rho \to 1$. Lasso: infinite ratio (one feature zeroed). Random forest: $O(1/\sqrt{T})$ convergence (the mechanism behind Dash).

Resolution. Dash: average SHAP values over ensemble of retrained models.

Experiments. 20 XGBoost models on UCI datasets, differing only in random seed. SHAP flip rate: 33%. Prediction equivalence: >99%. Four of twenty models achieved 72–78% held-out accuracy; all twenty are included in the analysis.

S2.2 Instance 2: Attention Maps

Explanation system. $\mathbf{\Theta}$: neural network weight configurations. Y: input–output functions. $\mathcal{H} = \Delta^T$: attention distributions over T tokens. $\perp$: different argmax tokens.

Rashomon property. D'Amour et al. [1] and Jain and Wallace [3] demonstrate that functionally equivalent networks attend maximally to different tokens.

Experiments. 10 DistilBERT models with weight perturbations ($\sigma \in \{0.01, 0.02\}$) on 200 SST-2 sentences. Prediction agreement: 100%. Argmax flip rate (weight perturbation): 60.0% [95% CI: 59–61%]. Under full retraining (20 models, last 2 transformer layers retrained from random initialization), the argmax flip rate is 19.9%. We report the conservative retraining figure in the main text. Mean Kendall τ: 0.30.

Resolution. Averaged attention rollout over the Rashomon set of equivalent models.

S2.3 Instance 3: Counterfactual Explanations

Explanation system. $\mathbf{\Theta}$: model parameters defining decision boundaries. Y: prediction vectors on test set. $\mathcal{H} = \mathcal{X}$: counterfactual points. $\perp$: opposite directions from query point ($(x'_1 - x_0)^\top (x'_2 - x_0) < 0$).

Rashomon property. Within a small loss tolerance, models can place decision boundaries on opposite sides of the query point [4].

Experiments. 20 XGBoost models on German Credit. Direction flip rate: 23.5% [CI: 20.6–26.6%]. Cross-model recourse validity: 61.3%. Most unstable feature: `checking_status` at 43% flip rate.

Resolution. Rashomon-robust recourse: the set of counterfactuals valid across all models in the Rashomon set [5].

S2.4 Instance 4: Concept Probes (TCAV)

Explanation system. $\mathbf{\Theta}$: weight configurations. Y: input–output functions. $\mathcal{H} = \mathbb{R}^c$: concept activation vectors (unit directions). $\perp$: structurally distinct directions not related by concept-subspace rotation.

Rashomon property. Bolukbasi et al. [6] and Ravfogel et al. [7] show that concept directions vary substantially across architectures and seeds.

Experiments. 15 MLP classifiers (128–64, ReLU) on `load_digits`. Prediction agreement: 97.2%. Mean $|\cos|$: 0.10 (nearly orthogonal). Concept direction instability ($1 - |\cos|$): 0.90. TCAV score SD: 0.16.

Resolution. Averaged concept activation vector or subspace over the Rashomon set.

S2.5 Instance 5: Causal Discovery

See main text, Section 3.2. The Rashomon property is derived from Markov equivalence. Resolution: CPDAG reporting [8].

Implementation note. The CI extraction function `ciFromDAG` is a simplified formula correct for chain and fork structures on three nodes. It does not implement full d-separation (in particular, it does not handle collider structures correctly). The Rashomon witness uses only chain and fork DAGs, for which the formula is verified by computation. A complete formalization of d-separation in Lean 4 is a substantial undertaking beyond the scope of this work.

S2.6 Instance 6: Model Selection

Explanation system. $\mathbf{\Theta}$: model classes (e.g., depth-3 tree, depth-5 tree, linear model). Y: prediction accuracy on test set. $\mathcal{H}$: model identities. $\perp$: different model choices.

Rashomon property. The Rashomon set generically contains multiple models with near-identical accuracy [4, 9].

Experiments. 20 XGBoost configurations on UCI datasets. Best-model flip rate: 80.0%. ΔAUC < 0.02.

Resolution. Ensemble achieves comparable performance (AUC 0.951 vs. 0.958 best-single) with evaluation-stable predictions across splits. Report the Rashomon set or ensemble rather than a single "best" model.

S2.7 Instance 7: Saliency Maps (GradCAM)

Explanation system. $\mathbf{\Theta}$: CNN weight configurations. Y: image classification function. $\mathcal{H}$: saliency heatmaps (spatial attribution maps). $\perp$: different peak-activation locations.

Rashomon property. Adebayo et al. [10] show that saliency maps can be insensitive to model weights, and Hooker et al. [11] demonstrate that saliency rankings shift under model perturbation.

Experiments. 10 ResNet-18 models with weight perturbations on CIFAR-10. Peak flip rate: 9.6%. Prediction agreement: 78.8%, reflecting sensitivity of ImageNet-pretrained ResNet-18 to perturbation on CIFAR-10.

Resolution. Averaged GradCAM heatmap over equivalent models.

S2.8 Instance 8: Token Citation Explanations

Explanation system. $\mathbf{\Theta}$: language model weight configurations. Y: next-token prediction function. $\mathcal{H}$: token citations—the top-k most-attended tokens extracted via attention rollout as a proxy for language model self-explanation. $\perp$: different cited evidence for the same prediction.

Rashomon property. Turpin et al. [12] show that chain-of-thought explanations are unfaithful: models produce different reasoning chains for the same answer. Lanham et al. [13] and Ye and Durrett [14] confirm instability across model variants.

Experiments. 10 DistilBERT models on SST-2. Citation flip rate: 34.5%. Prediction agreement: 100%.

Resolution. Averaged attention rollout or ensemble explanation aggregation.

S2.9 Instance 9: Mechanistic Interpretability

See main text, Section 3.3. Rashomon property: 85 valid circuits for a single task [15]. Resolution: circuit equivalence classes.

S3 Experiment Methodology

S3.1 General Protocol

For each instance, we follow a common experimental design:

1. **Train K models** from independent random seeds on the same dataset with the same architecture and hyperparameters.
2. **Verify functional equivalence**: confirm that all K models achieve statistically indistinguishable performance (prediction agreement $> 95\%$ or $\Delta\text{AUC} < 0.03$).
3. **Extract explanations**: apply the domain-specific explanation method to each model.
4. **Measure instability**: compute pairwise instability metrics across all $\binom{K}{2}$ model pairs.

5. **Negative control**: verify that instances without the Rashomon property (e.g., unique-solution linear regression) show 0% instability.
6. **Bootstrap confidence intervals**: 1000 bootstrap resamples for all reported metrics.

S3.2 Datasets

- **Attribution, counterfactual, model selection**: UCI German Credit, UCI Adult, and synthetic collinear datasets.
- **Attention, token citation**: Stanford Sentiment Treebank (SST-2), 200 sentences.
- **Concept probe**: `sklearn load_digits`, 8×8 digit images.
- **Saliency map**: CIFAR-10, 100 test images.

S3.3 Model Architectures

- **Attribution**: XGBoost (100 trees, max depth 6).
- **Attention / token citation**: DistilBERT (6 layers, 12 heads, 66M parameters) with Gaussian weight perturbation.
- **Counterfactual**: XGBoost on German Credit.
- **Concept probe**: MLP (128–64, ReLU), trained with SGD.
- **Model selection**: XGBoost with 20 hyperparameter configurations.
- **Saliency**: ResNet-18 with Gaussian weight perturbation.

S3.4 Perturbation Protocol for Attention and Saliency

For transformer and CNN models, we use weight perturbation within the zero-loss manifold rather than full retraining. Gaussian noise with standard deviation σ is added to all model weights. The perturbation magnitude σ is chosen to preserve prediction accuracy within 1% of the original model. This faithfully models the Rashomon set: the perturbed models are functionally equivalent but differ in internal representations.

Limitation: The distribution over perturbations may differ from the distribution induced by stochastic gradient descent. Full retraining experiments for transformers are computationally expensive and are left to future work.

S3.5 Multiple Comparisons

We report 95% bootstrap confidence intervals for each primary metric. Across seven experiments with multiple metrics, the total number of statistical comparisons is approximately 20. We do not apply formal multiple comparison correction (e.g., Bonferroni, Benjamini–Hochberg) for two reasons: (1) each experiment tests a domain-specific hypothesis that stands independently of the others, and (2) the primary question is not whether any single experiment reaches significance but whether the pattern of Rashomon-driven instability is consistent across domains. All positive results exceed the 20% practical significance threshold by substantial margins (range: 19.9%–80%), reducing the risk of false positives from multiple testing.

S4 Lean Verification Details

S4.1 Summary Statistics

Table S1: Lean 4 formalization summary.

Component	Count	Notes
Files	74	All in `UniversalImpossibility/`
Theorems + lemmas	349	Verified by `grep -c`
Axioms	72	Domain-specific; core uses 0
`sorry`	0	Zero admitted goals
Core impossibility	1 theorem	Zero axiom dependencies
GBDT instance	4 axioms	Surjectivity, first-mover, split structure
DASH resolution	6 axioms	+ cross-group symmetry
Causal instance	3 axioms	CPDAG Markov equivalence
Neural net instance	2 axioms	Overparameterization + symmetry
Fairness instance	2 axioms	Unequal base rates + equal accuracy

S4.2 Axiom Inventory

The 72 axioms fall into four categories:

Type declarations (6 axioms). `Model`, `numTrees`, `numTrees_pos`, `attribution`, `splitCount`, `firstMover`. These declare the types and basic constants of the GBDT framework.

Core behavioral properties (6 axioms). `firstMover_surjective`, `splitCount_firstMover`, `splitCount_nonFirstMover`, `proportionality_global`, `splitCount_crossGroup_symmetric`, `splitCount_cross` These encode the structure of gradient-boosted decision trees.

Measure infrastructure (2 axioms). `modelMeasurableSpace`, `modelMeasure`. These connect the model space to Mathlib's measure theory library.

Universal instance axioms (58 axioms). Per-instance type and Rashomon witness axioms for the nine instances (attention, counterfactual, concept, causal, model selection, saliency, token citation, mechanistic interpretability, fairness).

S4.3 Axiom Stratification

The axiom dependencies are strictly stratified:

- **Core impossibility** (`explanation_impossibility`): **0 axioms**. The Rashomon property enters as a hypothesis, not an axiom.
- **Attribution impossibility** (`attribution_impossibility`): **0 behavioral axioms** (only type declarations).

- **GBDT impossibility** (`gbdt_impossibility_local`): 4 axioms (surjectivity, first-mover, non-first-mover, split-count).
- **Quantitative bounds** (`impossibility`): 5 axioms (+ proportionality).
- **DASH resolution** (`consensus_equity`): 6 axioms (+ cross-group symmetry).

S4.4 Axiom Substitution Analysis

We have verified that:

1. Removing any single core behavioral axiom blocks the corresponding impossibility proof (each axiom is necessary).
2. The Rashomon property can be substituted with any domain-specific witness (Markov equivalence, collinearity, weight-space symmetry) without changing the core proof.
3. The causal instance achieves *zero* axioms for the Rashomon property itself (it is derived from Markov equivalence as a theorem).

S4.5 Key Theorem List

Theorem name	File	Axioms
`explanation_impossibility`	ExplanationSystem.lean	0
`attribution_impossibility`	Trilemma.lean	0
`attribution_impossibility_weak`	Trilemma.lean	0
`attribution_impossibility_abstract`	AttributionInstance.lean	0
`attention_impossibility`	AttentionInstance.lean	3
`counterfactual_impossibility`	CounterfactualInstance.lean	3
`concept_impossibility`	ConceptInstance.lean	3
`causal_instance_impossibility`	CausalInstance.lean	3
`mech_interp_impossibility`	MechInterpInstance.lean	3
`gInvariant_stable`	UniversalResolution.lean	0
`universal_design_space_dichotomy`	UniversalDesignSpace.lean	0
`no_rashomon_all_three`	Necessity.lean	0
`generic_underspecification`	Ubiquity.lean	0
`consensus_equity`	Corollary.lean	6
`gbdt_impossibility_local`	General.lean	4
`ratio_tendsto_atTop`	Ratio.lean	5
`lasso_impossibility`	Lasso.lean	2
`causal_rashomon_from_markov`	MarkovEquivalence.lean	0

S5 Practitioner Decision Guide

Table S3 provides actionable guidance. Each row corresponds to a common practitioner goal and specifies which desideratum is sacrificed, what method class achieves the remaining two, and example tools.

Table S3: Practitioner decision table (Table S2): which desideratum to sacrifice for each goal.

Your goal	Sacrifice	Method	Example tools
Debug THIS model	Stability	Model's own explanation	SHAP, GradCAM
Monitor across retrains	Decisiveness	Aggregate over Rashomon set	Dash, CPDAG
Regulatory template	Faithfulness	Fixed rule-based explanation	Predefined feature lists
Safety audit	Decisiveness	Conservative bounds	Worst-case over Rashomon set

S6 Regulatory Analysis

S6.1 EU AI Act Article 13 (Transparency)

The EU AI Act [16] requires "appropriate transparency" and "meaningful information about the logic involved" for high-risk systems. "Meaningful" implies faithfulness and decisiveness; "appropriate" across model versions implies stability. The main theorem shows all three simultaneously is impossible under the Rashomon property.

Resolution. Report G-invariant explanations with explicit uncertainty. Disclose which property is sacrificed (typically decisiveness).

S6.2 NIST AI RMF (MAP 1.5, MEASURE 2.6)

The NIST framework [17] requires interpretability to be "defined, appropriate to the AI risk, and documented." This implicitly assumes all three desiderata are simultaneously achievable.

Resolution. Explicitly specify which leg of the trilemma is prioritized for each use case and document the tradeoff in the system card.

S6.3 OCC SR 11-7 (Model Risk Management)

SR 11-7 [18] requires conceptual soundness (faithfulness), outcomes analysis across versions (stability), and effective challenge (decisiveness). These three pillars map directly onto the trilemma.

Resolution. Use ensemble explanations (Dash) for stability and partial faithfulness. Document decisiveness limitations. Direct effective challenge at modeling choices rather than explanation artifacts.

S7 Cross-Domain Connections

S7.1 Quine–Duhem Underdetermination

Quine [19] argued that "any statement can be held true come what may, if we make drastic enough adjustments elsewhere in the system." Duhem [20] held that no physical hypothesis can be tested in isolation. The main theorem formalizes this: the configuration space is the space of theories, the observable space is empirical data, and the Rashomon property captures the existence of empirically adequate theories with incompatible explanations.

S7.2 Economics: Partial Identification

Manski [21] developed partial identification theory for settings where observational data do not pin down a unique model. The identified set in Manski's framework corresponds to an observe-fiber in ours; partial identification is the economic instantiation of the Rashomon property.

S7.3 Database Theory: View Update Problem

Bancilhon and Spyratos [22] proved that translating updates on a database view back to the base tables is inherently ambiguous when the view is not injective. This is precisely our framework: the base tables are configurations, the view is the observation map, and the ambiguity is the Rashomon property.

S7.4 Medical Diagnosis: Differential Diagnosis

When multiple conditions produce the same symptoms (the observable), diagnosis faces the trilemma. Differential diagnosis—the standard medical practice of listing all conditions consistent with the symptoms—is precisely G-invariant resolution: it sacrifices decisiveness (single diagnosis) for faithfulness and stability.

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